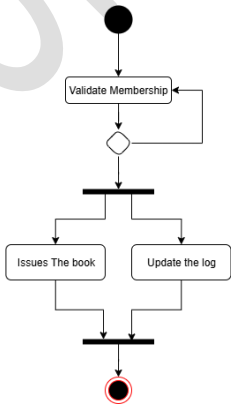
SE – Mid Exam – Mock Questions (Student Made)

Select all that apply

1. Which of the following are true about SDLC?
    a. Feasibility study should be performed before the SDLC
    b. The high-level design provides a comprehensive and in-depth blueprint of the software along with the detailed logic and components
    c. SRS (Software Requirement Specification) is the output at the end of the design phase
    d. For quick prototypes and scripts SDLC is efficient, rather than programming.
    e. Writing the code according to the design is known as the deployment
2. Which of the following are feasibility types?
    a. Logical Feasibility
    b. Digital Feasibility
    c. Technical Feasibility
    d. Schedule Feasibility
    e. Authentication Feasibility
3. Which of the following are true about requirement types?
    a. Functional requirements describe what system should do
    b. Non-functional requirements are “nice to have” type features, often focusing on efficiency of the system
    c. Limitations of the system are always unintentional and should be avoided
    d. In an ATM machine software, withdrawing cash is a non-functional requirement
    e. The system design should only focus on fulfilling functional requirements
4. Consider the following activity diagram and select the correct statements.



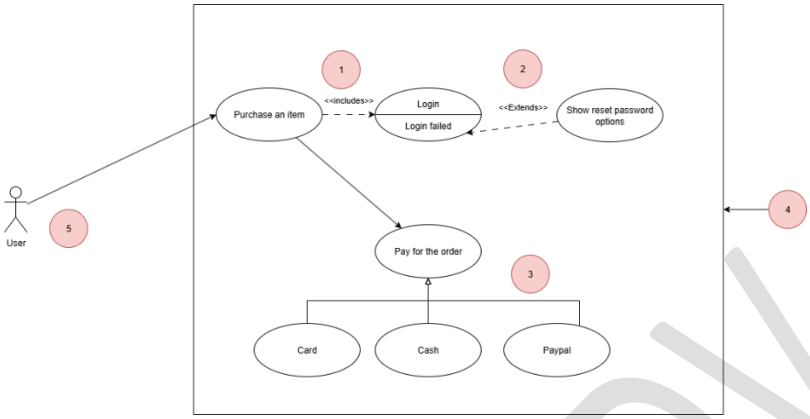
    a. The fork is used to represent a dividend of a single flow into multiple flows.
    b. The fork is used to represent a dividend of a single path into multiple paths.
    c. The decision node and expression for membership validation are correct in the diagram.
    d. The joint can be replaced with a merge node.
    e. The joint must be followed by a fork.

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5. Select the correct statements about activity diagrams:
    a. A rake symbol emphasizes undefined number of iterations
    b. The asterisk mark can be replaced with a decision node.
    c. In a loop, there is no need to mention the guard conditions.
    d. In a fork, the control flows through all forked paths.
    e. A sub-activity is indicated by an empty circle.
6. Consider a Use Case Diagram for an online banking system. A central use case is "Make a Transaction." Any transaction, whether it is a "Pay Bill" or "Transfer Funds," must first "Authenticate User." If a transaction is over $10,000, it optionally triggers a "Notify via SMS" action. "Pay Bill" and "Transfer Funds" are specific types of transactions.
    a. The relationship between "Authenticate User" and "Make a Transaction" should be modeled with an <<extend>>, as authentication is an extension of the transaction process.
    b. "Pay Bill" and "Transfer Funds" should have a **Generalization** relationship with "Make a Transaction," where "Make a Transaction" is the parent use case.
    c. The relationship between "Notify via SMS" and "Make a Transaction" should be modeled with <<include>>, as it is included in the core process.
    d. The <<include>> relationship is appropriate for "Authenticate User" because it represents mandatory, non-standalone functionality that is reused across multiple parent use cases.
    e. The <<extend>> relationship for "Notify via SMS" is appropriate because the base use case ("Make a Transaction") can complete successfully without it, and it only runs under a specific condition.
7. team is defining requirements for a university library system. Which of the following are valid examples of their respective requirement types?
    a. **Functional:** The system shall allow a librarian to add a new book to the catalog.
    b. **Non-Functional:** The system's search results must be displayed to the user within 2 seconds.
    c. **Constraint:** The system must be developed using the Java programming language.
    d. **Functional:** The user interface must be intuitive and easy for first-time users to navigate.
    e. **Non-Functional:** The system must be able to calculate fines for overdue books.
8. A manager wants to model the flow of handling an insurance claim. Which UML Activity Diagram elements would be correctly used for the following steps?
    a. A **Decision Node** to check if the claim amount is "Above $5,000" or "Below $5,000".
    b. An **Initial Node** (filled circle) to signify the start of the process when a claim is first received.
    c. A **Fork** to send the claim to both the "Fraud Detection Unit" and the "Damage Assessment Team" to be processed concurrently.
    d. A **Merge Node** to bring the concurrent flows from the "Fraud Detection Unit" and "Damage Assessment Team" back into one.
    e. A **Sub-activity** (rake symbol) for the "Assess Damage" action, indicating it is a complex process detailed in a separate activity diagram.

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Consider the following diagram and answer the questions.



9. Select the correct statements
   a. Users must log in before purchasing an item
   b. Users need to reset their password when purchasing an item.
   c. Users need to reset their password only if login is failed.
   d. Login is optional in order to purchase an item
   e. Users will pay for an item only after a successful login
10. Select **the** correct relationship/ entities names for 1,2,3,4 and 5.
    a. 1: Include Relationship
       2. Extend Relationship
       3. System Boundary
       4. Digital Boundary
       5. Actor

    b. 1. Aggregation
       2. Dependency
       3. Generalization
       4. System Boundary
       5. Actor

    c. 1. Aggregation
       2. Dependency
       3. Inheritance
       4. Machine Boundary
       5. Actor

    d. 1. Include Relationship
       2. Extend Relationship
       3. Generalization Relationship
       4. System Boundary
       5. Actor

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e. 1. Dependency
   2. Aggregation
   3. Inheritance
   4. System Boundary
   5. Actor

Select the most suitable answer

11. In use case diagrams, preconditions can be often shown as:
    a. Include Relationships
    b. Extend Relationships
    c. Exclude Relationships
    d. Generalization Relationships
    e. Aggregation Relationships
12. Swim-lanes in an activity diagram are used to:
    a. Representing different actors
    b. Representing different paths
    c. Representing different conditions
    d. Representing different flows
    e. Representing different activity partitions
13. The diamond shaped notations in activity diagrams with no guard conditions are known as:
    a. Merge Notation
    b. Joint Notation
    c. Fork Notation
    d. Loop Notation
    e. Decision Notation
14. Which of the following is **not** an agile methodology?
    a. TDD
    b. XP
    c. SDLC
    d. Kanban
    e. Scrum
15. Select the role of Scrum master in Scrum
    a. Decides which features and functionalities to build
    b. Maintain communication between all participants.
    c. Represent the client.
    d. Taking the exert control over the team
    e. Removing the impediments
16. Select the SCRUM artifact
    a. Product backlog
    b. Sprint Planning
    c. Daily Scrum
    d. Sprint Review
    e. Sprint Retrospective

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17. Which of the following is not a quadrant of the Spiral Model?
    a. Cumulative Cost
    b. Evaluate alternatives and Identify Risks
    c. Develop, verify next level product
    d. Plan next phases
    e. Determine objectives, alternatives and constraints
18. Which of the following models use a linear approach to solve a problem.
    a. Waterfall Model
    b. Spiral Model
    c. Scrum Framework (Agile)
    d. Iterative Waterfall
    e. Incremental Model

Answer the question 19 - 21 using the following description

A team has decided to follow SCRUM framework for a software product. They listed down the **prioritized list of works** and there appears to be 150 of them. The project needs to be completed in 5 days. They completed it as follows.

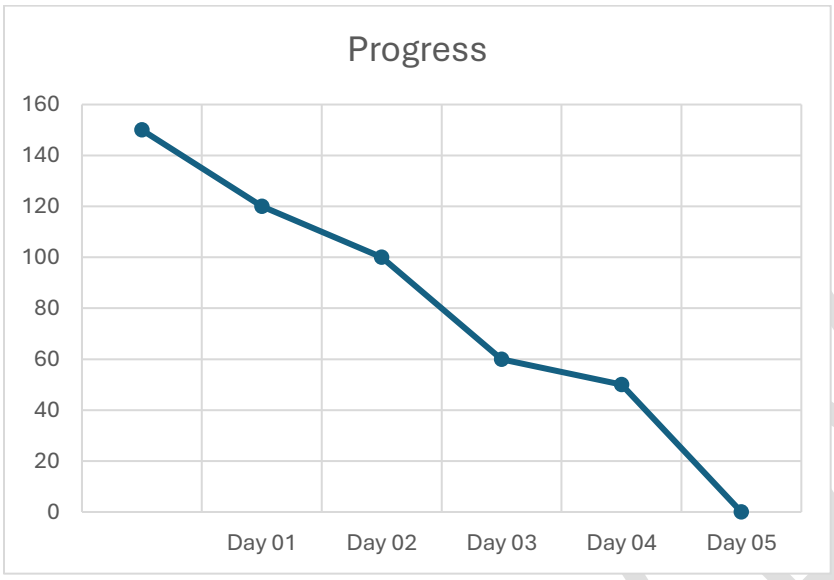
Day 01: 30 tasks
Day 02: 20 tasks
Day 03: 40 tasks
Day 04: 10 tasks
Day 05: 50 tasks

19. What is the ideal velocity for the burn down chart?
    a. 30
    b. 5
    c. 750
    d. 20
    e. 3

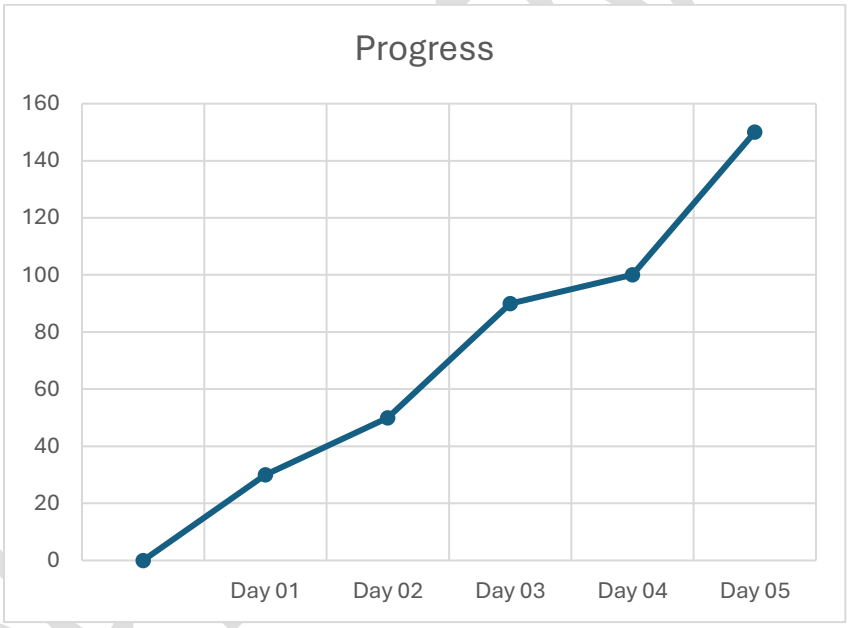
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20. Select the actual burndown chart for the above scenario

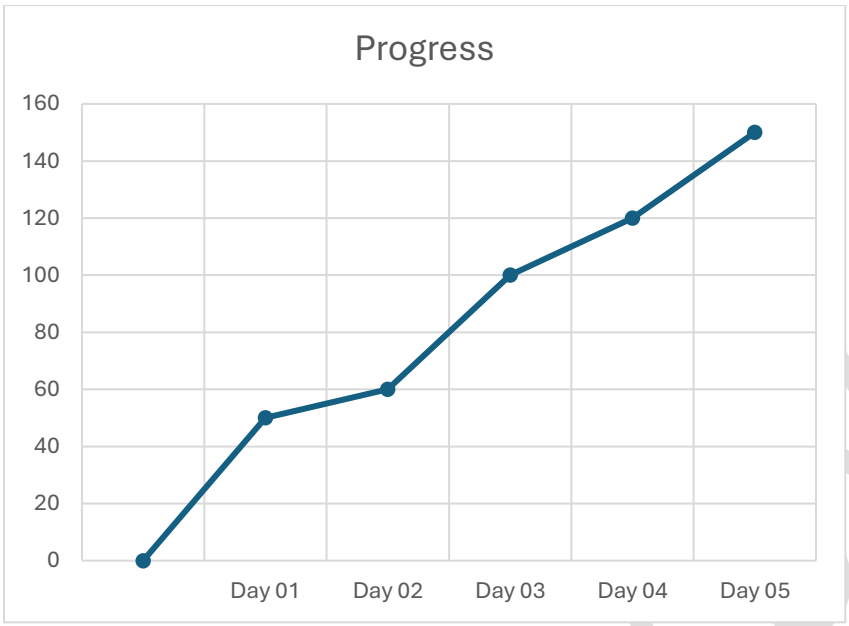
a.



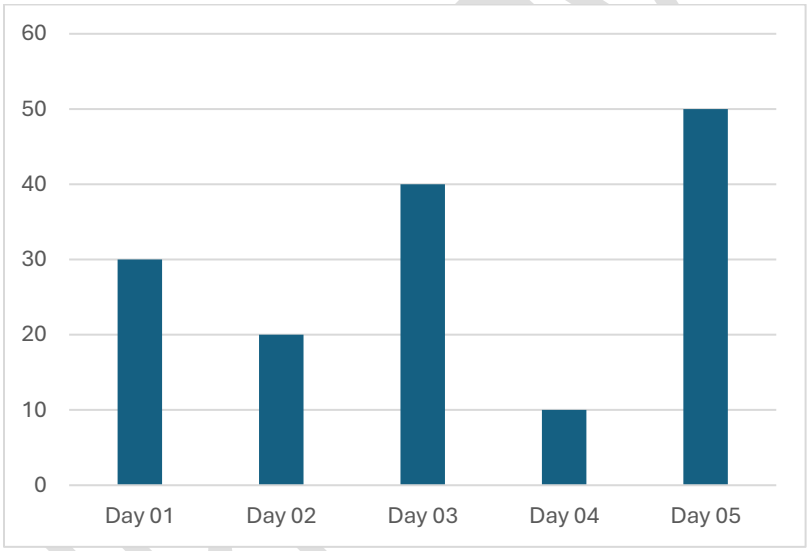
b.

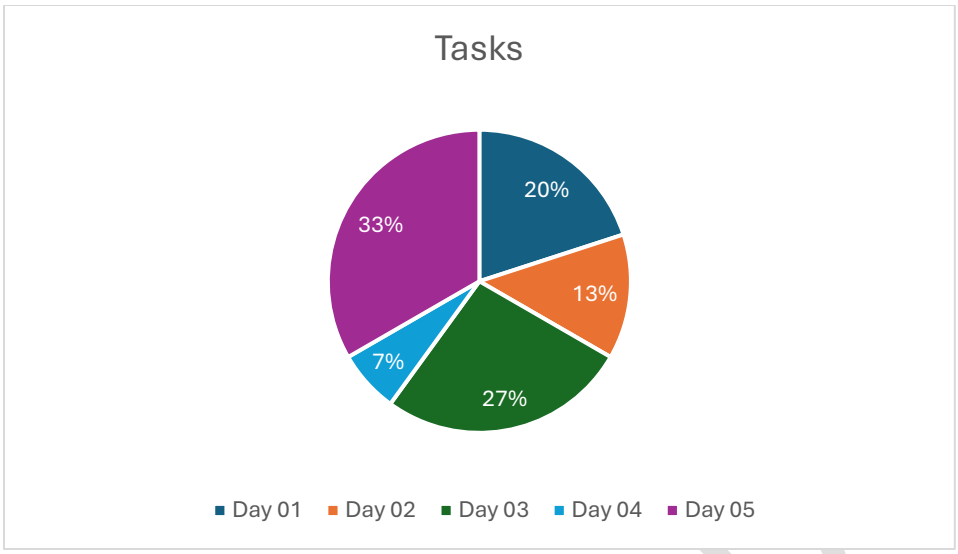


c.


Progress
160
140
120
100
80
60
40
20
0
Day 01
Day 02
Day 03
Day 04
Day 05

d.


60
50
40
30
20
10
0
Day 01
Day 02
Day 03
Day 04
Day 05

e. 

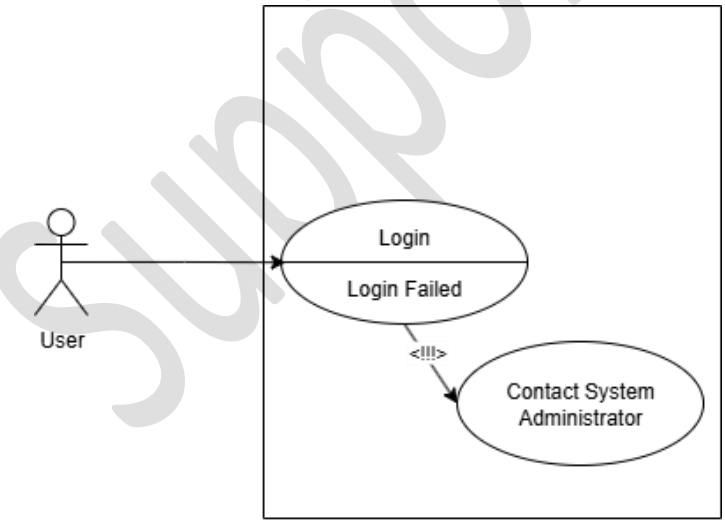
21. In this project they held a meeting for every single day for all 5 days at 7:00 P.M. This meeting is known as:
    a. Sprint backlog
    b. Product Backlog
    c. Ideal Meeting Velocity
    d. Momentum Meeting
    e. Daily Scrum
22. In product backlog the highest priority tasks appear in the:
    a. Top of the product backlog
    b. Middle of the product backlog
    c. Bottom of the product backlog
    d. Spread around everywhere evenly
    e. Can be appear anywhere
23. A university has software as their LMS (Learning Management System) and they recently upgraded from windows 2016 server to windows 2025 server. The software was updated due to incompatibility with the new server. These kinds of maintenances are known as:
    a. Feature maintenance
    b. Corrective maintenance
    c. Adaptive maintenance
    d. Perfective maintenance
    e. Cumulative maintenance
24. Which one of the following is **not** a key ethical value in Software Engineering?
    a. Piracy
    b. Transparency
    c. Fairness
    d. Responsibility
    e. Privacy

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25. A mobile software needs to be delivered within one month and the requirements may change while developing the app. The project team members are structured and not flexible. Which of the following is the best model/ framework, considering given information.
    a. SCRUM
    b. Kanban
    c. Waterfall
    d. Iterative Waterfall
    e. Spiral
26. In incremental model, a small part developed step by step is known as:
    a. An increment
    b. A sprint
    c. A SCRUM
    d. A partition
    e. Prototype
27. What is the corresponding validation phase step to the **software design** in V-Model?
    a. System Testing
    b. Unit Testing
    c. Acceptance Testing
    d. Prototype Testing
    e. Integration System
28. Imagine you are building an automated driving system for an electronic vehicle. If you want to catch errors as soon as each phase begins, which of the following is the best model for that task?
    a. V-Model
    b. Iterative Waterfall Model
    c. Prototyping
    d. Incremental Model
    e. Spiral Model
29. Which of the following is **not** a part of **spiral model loop**?
    a. Planning
    b. Engineering
    c. Burn Down Charts
    d. Evaluation
    e. Risk Analysis
30. In use case diagrams communication path between an actor and a use case it participates in known as:
    a. Dependency
    b. Generalization
    c. Association
    d. Include
    e. Extend

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31. In a banking system manager can view the transactions done by each counter using an application on his office desktop PC. The application is also capable of viewing other bank contact details and generating reports, etc. Before doing any of that the manager must log into the system by using a fingerprint scanner. However, if the fingerprint is not recognized within three attempts, the security will be informed. Considering the **viewing transaction** option, what is the corresponding trigger to that use case?
    a. Manager selects to view the transactions
    b. Manager log into the system
    c. The fingerprint mismatch alert
    d. Informing the security
    e. Manager turns on the PC
32. While creating a use case scenario for a book borrowing system for a library a software engineer notices that the number of maximum that could borrow per a member is not defined. In which use case scenario table section he should mention it?
    a. Trigger
    b. Secondary Actors
    c. Extensions
    d. Main Scenario
    e. Open Issues
33. What is **not** a main component of a case diagram?
    a. Actors
    b. SCRUM master
    c. System
    d. Use Cases
    e. Links
34. The <!!!> should be replaced with:



    a. <<extend>>
    b. <<include>>
    c. <<forward>>
    d. <<else>>
    e. <<If failed>>

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35. The joint represents the join of <<**01**>> while merge node represents join of **<<02>>**.
    a. 01. Flows
       02. Paths

    b. 01. Gesture
       02. Logics

    c. 01. Paths
       02. Flows

    d. 01. Decisions
       02. Tasks

    e. 01. Forks
       02. Decisions

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